

Track 49: Industrial Safety

June 2026

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Tier 3 — Specialized | Departments: All Engineering Branches

Skill path: Safety Regulations → Hazard Identification → Risk Assessment → PPE → LOTO → Fire Safety → OSHA/Factory Act → Safety Audits

Related: [Track 25 — PLC & Industrial Automation](#) · [Department Map](#)

Track Overview

- Who this is for:** All engineering students and shop-floor professionals; mandatory add-on for manufacturing, EEE, Mechatronics, and Civil site roles.
- Prerequisites:** None; applicable across all departments.
- Tools:** HIRA templates, JSA forms, safety audit checklists, case study library, PPE demonstration kits.
- Certification targets:** OSHA 30-hour (awareness), NEBOSH IGC intro (6-month, optional)

Outcomes by Duration

Duration	Hours	Job Readiness	Key Deliverables
2 Weeks	40–50	Safety awareness	HIRA for sample workplace
1 Month	80–100	Safety officer basics	Complete safety management plan
3 Months	250–300	EHS intern ready	Audit report + training materials
6 Months	500–600	EHS coordinator track	Industry safety capstone

2-Week Crash Course

Days	Topics	Deliverable
D1–2	Safety culture, Factory Act/OSHA basics, rights/duties	Safety policy draft
D3–4	Hazard identification, HIRA, JSA	HIRA document
D5–6	PPE, machine guarding, electrical safety	PPE assessment checklist
D7–8	Fire safety, emergency response, first aid basics	Emergency plan
D9–10	Capstone	Workplace safety audit report

1-Month Foundation

Week	Topics	Project
W1	LOTO, confined spaces, hot work permits	Permit system documentation
W2	Chemical safety, MSDS, spill response	Chemical handling SOP
W3	Ergonomics, noise, ventilation, lighting	Workplace assessment report
W4	Incident investigation, near-miss reporting, capstone	Safety management system binder

Department-Specific Modules (Elective Deep-Dive)

Department	Focus Module
EEE / Mechatronics	Electrical arc flash, LOTO for drives, PLC safety circuits
Mechanical / CNC	Machine guarding, CNC safety, material handling
Civil	Construction site safety, scaffolding, excavation, fall protection
ECE / IoT	Lab electrical safety, battery storage safety
Chemical	Process safety, HAZOP intro (stretch)

3–6 Month Progression

Safety training delivery + mock audit + ISO 45001 awareness + capstone safety improvement project

at partner industry (6-month).

Assessment Rubric

Weekly Quiz 20% | Audit/checklist labs 40% | Project 30% | Portfolio 10%

Appendix: Mentor Notes

Cross-department Tier 3 course. Recommend as add-on to Tier 2 manufacturing tracks (PLC, Mechatronics, CNC). Include at least one site visit or simulated audit.